INCH POUND

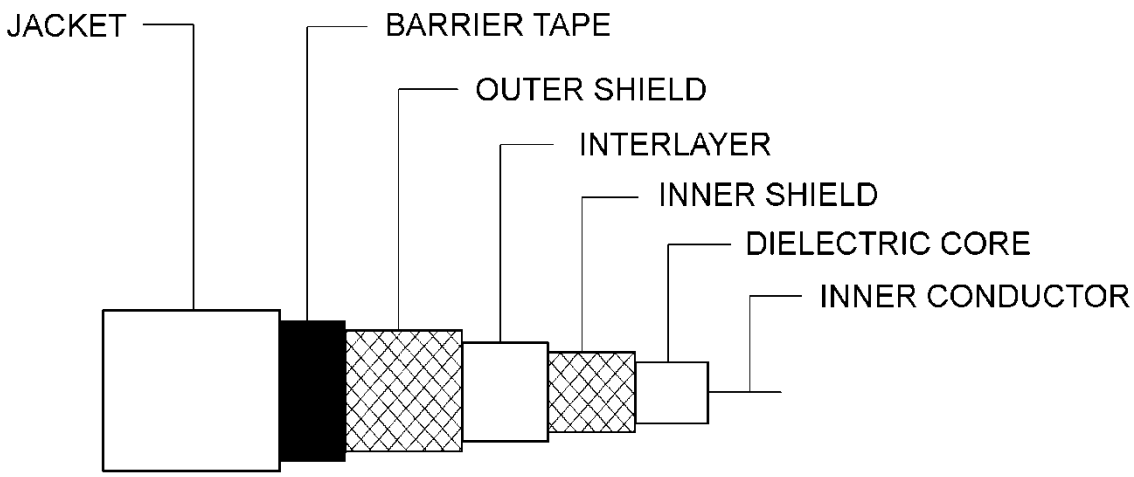
MIL-DTL-17/135E
22 April 2014
SUPERSEDING
MIL-DTL-17/135D
03 June 2013

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, TRIAXIAL, .500 INCH, 50 OHM, WATER BLOCKED, NON-WATER BLOCKED AND CROSSLINKED, LOW SMOKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.



NOTE: Cable should be constructed of non-toxic material.

FIGURE 1. General configuration.

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TABLE I. Description.

Component	Construction
Inner conductor (M17/135-00001 and -00002)	Seven strands of silver-coated copper wire each strand .0296 inch diameter. Diameter: .089 inch ±0.001
Inner conductor(M17/135-00003 thru -00006)	Solid silver-coated copper wire diameter. Diameter: .081 inch ±0.001
Dielectric core	Type A-1: Solid polyethylene Diameter: .285 inch ± 0.010
Inner shield	Single braid of AWG No. 33 silver-coated copper wire Diameter: .340 inch maximum Coverage: 90%, minimum Carriers: 24 Ends: 7 Picks/inch: 8 Separator tape optional
Interlayer (M17/135-00001 thru M17/135-00004)	Type A-1: Solid polyethylene Diameter: .365 inch ± .010
Interlayer (M17/135-00005 and M17/135-00006)	Cross-linked polyolefin Diameter: .365 inch ± .010
Outer shield	Single braid of AWG No. 33 silver-coated copper wire Diameter: .414 inch, maximum Coverage: 85%, minimum Carriers: 24 Ends: 8 Picks/inch: 5.11 Separator tape optional
Jacket M17/135-00005 1/ and M17/135-00006	Cross-linked polyolefin Diameter: .500 inch ± .010, maximum
Jacket M17/135-00001 1/ and M17/135-00002	Type XIII: Polyurthane Diameter: .500 inch ± .010, maximum
Jacket M17/135-00003 1/ and M17/135-00004	Type IIIa: Polyethylene Diameter: .500 inch ± .010

1/ Free strippable blocking compound needed where necessary to meet hydrostatic test. The inner shield may be partially imbedded in the dielectric. Applicable to water blocked cables only.

ENGINEERING INFORMATION

Continuous working voltage: 3,700 V rms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

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Operating temperature range:

-40°C to +70°C (M17/135-00001 thru M17/135-00004)
-30°C to +85°C (M17/135-00005 and M17/135-00006)

Inner conductor properties:

DC resistance (maximum at 20°C): 1.62 ohms per 100 feet. (0.173 ohm per 100 feet for M17/135-00001 and M17/135-00002.)

Elongation: 25 percent, minimum. (30 percent minimum for M17/135-00001 and M17/135-00002)

Tensile strength: Not applicable.

Engineering note: This cable useful for non-hosing low temperature applications.

REQUIREMENTS

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

M17/135-00001,M17/135-00003,and M17/135-00005: 7 pounds minimum, 65 pounds maximum.
M17/135-00002,M17/135-00004,and M17/135-00006: 7 pounds minimum, 50 pounds maximum

Aging stability:

+90°C ± 2°C (M17/135-00003 and M17/135-00004)
+98°C ± 2°C (M17/135-00001, M17/135-00002, M17/135-00005 and M17/135-00006)

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend:

-55°C ± 2°C (M17/135-00001 thru M17/135-00004)
-30°C ± 2°C (M17/135-00005 and M17/135-00006)

Dimensional stability: +85°C ± 2°C.

Inner conductor from core: .125 inch, maximum.

Inner conductor from jacket: .250 inch, maximum.

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Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Flame propagation: Applicable to M17/135-00005 and M17/135-00006.

Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum. Applicable to M17/135-00005 and M17/135-00006.

Immersion test: Applicable to M17/135-00005 and M17/135-00006.

Tensile strength, percent of unaged minimum: 50. Applicable to M17/135-00005 and M17/135-00006.

Elongation, percent of unaged minimum: 50. Applicable to M17/135-00005 and M17/135-00006.

Smoke index: 25 maximum. Applicable to M17/135-00005 and M17/135-00006.

Toxicity index: 5 maximum. Applicable to M17/135-00005 and M17/135-00006.

Durometer hardness: (type A) 80 minimum. Applicable to M17/135-00005 and M17/135-00006.

Weathering: Applicable to M17/135-00005 and M17/135-00006.

Abrasion resistance: 75 cycles minimum (jacket only). Applicable to M17/135-00005 and M17/135-00006.

Tear strength: 35 pounds per inch minimum. Applicable to M17/135-00005 and M17/135-00006.

Heat distortion: 30 percent maximum distortion. Applicable to M17/135-00005 and M17/135-00006.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum. Applicable to M17/135-00005 and M17/135-00006.

Elongation: 160 percent, minimum. Applicable to M17/135-00005 and M17/135-00006.

Physical tests on aged jacket: Applicable to M17/135-00005 and M17/135-00006.

Air oven:

Tensile strength, percent minimum: 60.

Elongation, percent minimum: 60.

Hot oil immersion: Applicable to M17/135-00005 and M17/135-00006.

Tensile strength, percent minimum: 50.

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Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent, minimum. Applicable to M17/135-00005 and M17/135-00006.

Weight: 16 pounds per 100 feet, max (M17/135-00001 and –00002).
18.5 pounds per 100 feet, max (M17/135-00003 thru –00006).

Watertightness (not applicable to M17/135-00002, M17/135-00004, and M17/135-00006): One end of a five foot length of completed cable shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a water pressure of 25 lb_f/in^2 for a period of 6 hours. Any evidence of water leakage shall be cause for rejection.

Hydrostatic (open end) (not applicable to M17/135-00002, M17/135-00004, and M17/135-00006): A 5-foot specimen of completed cable shall be fitted into a gland similar to USL 44500B-12. The cable shall be capable of withstanding 1,000 lb_f/in^2 applied to the gland end of the specimen for a period of 2 hours. Any evidence of water leakage shall be cause for rejection.

Electrical:

Continuity: Applicable.

Spark test: 8,000 V rms, +10%, -0%.

Voltage withstanding: 10,000 V rms, minimum applied between the inner and outer conductors with the outer conductor grounded; 500 V dc minimum applied between the outer conductor and the extra shield with the extra shield grounded.

Insulation resistance: Not applicable.

Corona extinction voltage: 5,000 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

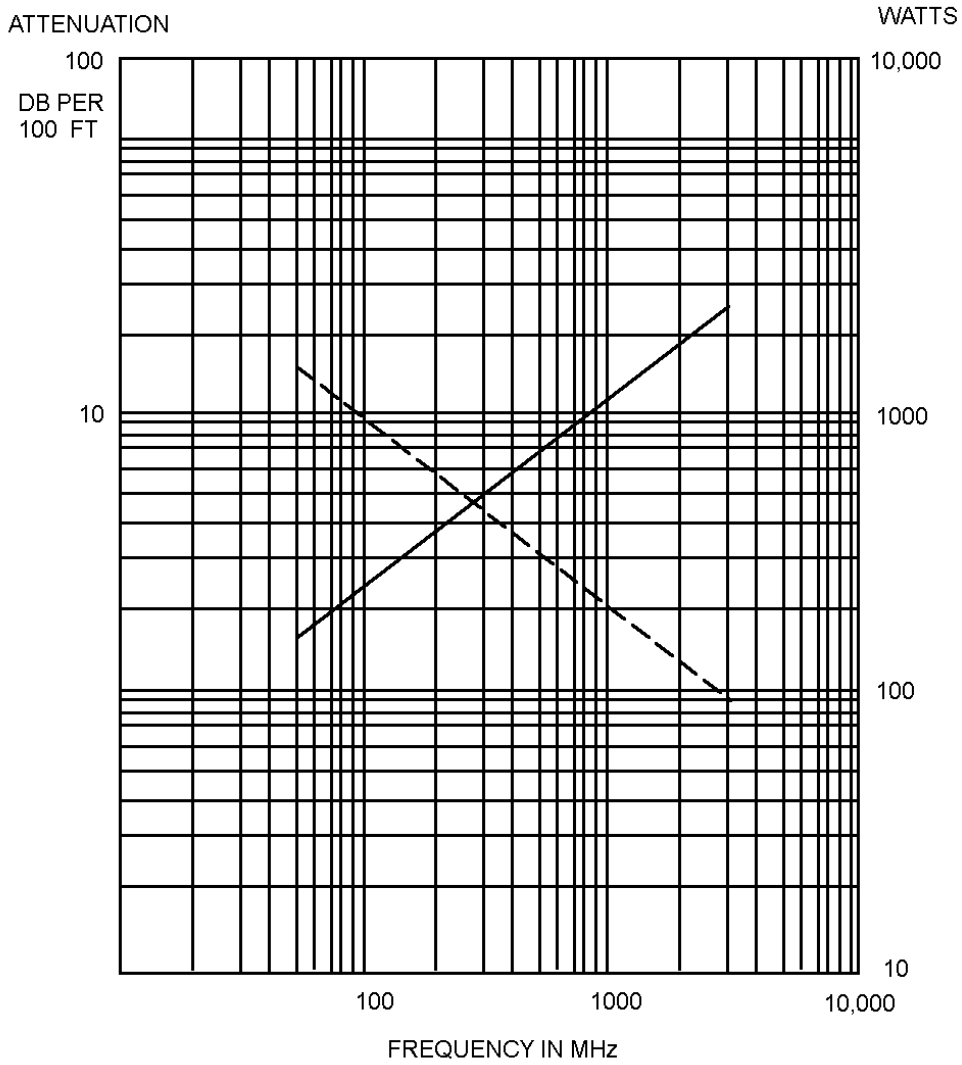
Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

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Part or Identifying Number (PIN): See table II.

Supersession data: See table II.



NOTE:
1. Maximum power (at 25°C seal level)
2. Maximum attenuation ———————
3. Test requirements shall be noted as line indicated on graph.

Frequency MHz	Attenuation dB	Watts dB
50	1.7	1400
100	2.6	850
400	6.4	350
1000	11.4	190
3000	23	90

FIGURE 2. Power rating and attenuation.

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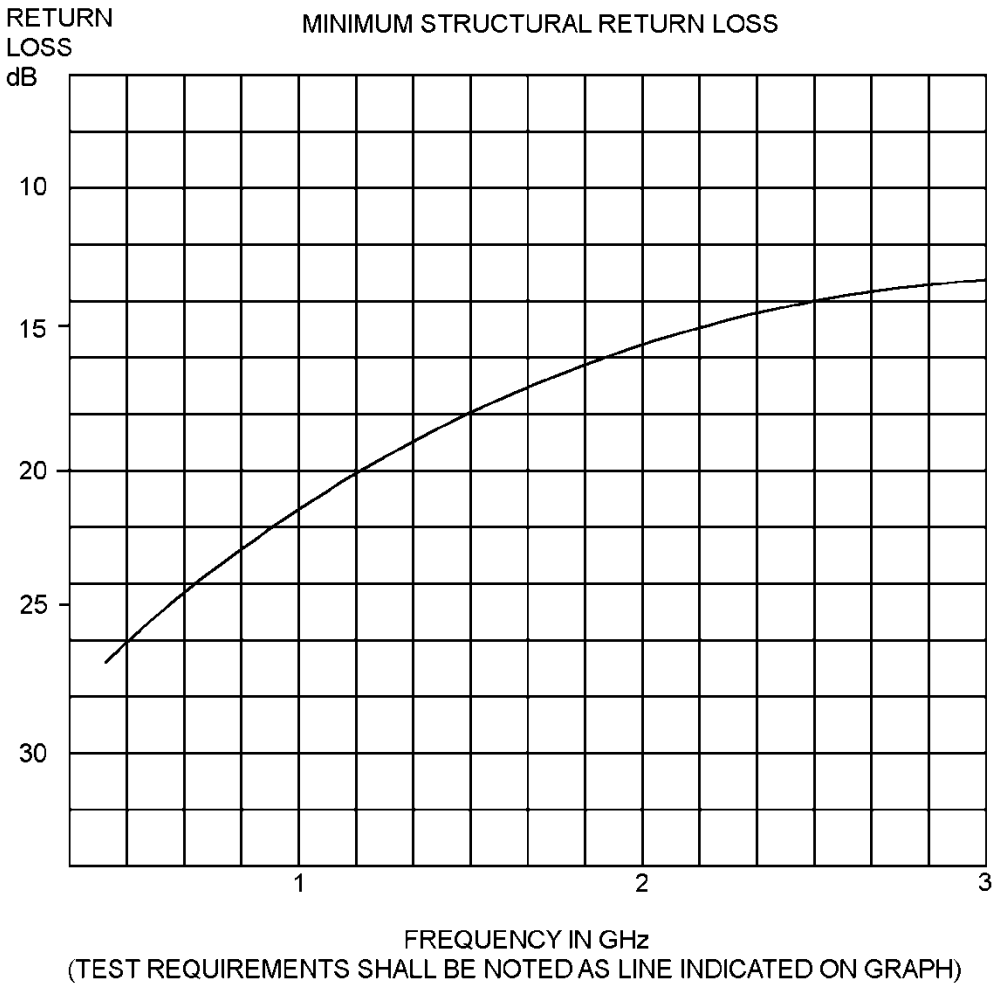


FIGURE 3. Structural return loss.

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SWR	Reflection coefficient	Return loss dB	SWR	Reflection coefficient	Return loss dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

Frequency MHz	MIN. SRL
50	27
1000	21
2000	16
3000	13.8

FIGURE 3. Structural return loss - Continued.

TABLE II. Cross-reference of PIN.

PIN	Superseded PIN type or designation	Type
M17/135—00001 1/	AA-3833	Water blocked
M17/135—00002 1/	TRF-8	Non-water blocked
M17/135—00003	----------	Water blocked
M17/135—00004	----------	Non-water blocked
M17/135—00005 2/	AA-7559	Water blocked
M17/135—00006 2/	AA-7560	Non-water blocked

1/ Inactive for new design.
2/ Cable previously manufactured using the barrier tape is acceptable for Government use until stock is purged.

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Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity
DLA - CC

(Project 6145-2014-030)

Review activities:
Army - AR, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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